



POWER BI
TURN DATA INTO INSIGHTS

Patient Analytics Dashboard



Healthcare Performance • Trends • Insights



DATE

30 April 2026



AUTHOR

Mohammed Abrar



VERSION

1.0



CONFIDENTIAL • INTERNAL USE ONLY

TABLE OF CONTENTS

Power BI Technical Documentation — Patient Analytics Dashboard

1. Report Theme		
Canvas Background		2
Color Palette		2
Typography		2
2. Data Sources		
Loaded Tables		3
Created / Reference Tables		3
Calendar Table (DimDate)		3
Blood Group Description Table		3
Insurance Description Table		4
Source of Awareness Table		4
3. Data Transformations		
Branches Table		5
Countries Table		5
Patients Table (Fact Table)		5
Gender Column — Value Replacement		5
Billing Type Column — Value Replacement		6
ResidencyType Column — Value Replacement		6
Calculated Column: Age		6
Calculated Column: Age Group		6
4. Data Modelling		
5. Key Performance Indicators (Measures)		
6. Report Pages & Visualizations		
Patient Demographic Insights Dashboard		9
Dashboard Header (Text Card)		9
Slicers		9
KPI Card		9
Charts		10
Patients Registration Trend Dashboard		12
Dashboard Header (Text Card)		12
Slicers		12

Charts	 13
--------	----------

1. Report Theme

1.1 Canvas Background

All report pages use a canvas background color of #DEECF9 (Canvas Blue) applied in the Power BI Desktop page settings under View → Page background. This provides a clean, clinical look consistent with healthcare analytics.

1.2 Color Palette

The following color palette is applied consistently across all visual elements — charts, slicers, cards, and table formatting — to maintain brand identity and visual hierarchy throughout the report.

Swatch	Hex Code	Color Name
	#DEECF9	Canvas Blue
	#16697A	Deep Teal
	#489FB5	Mid Blue
	#82C0CC	Sky Blue
	#EDE7E3	Warm Off-White
	#FFA62B	Amber / Gold
	#CC704B	Rust / Burnt Orange

1.3 Typography

Element	Font Family	Size & Style
Dashboard Title	Segoe UI	18pt, Black, Bold
Chart Title	DIN	14pt, Black, Bold Italic
Chart Subtitle	Segoe UI	9pt, Black
Slicer Header	DIN	11pt, Black, Bold
Slicer Values	Segoe UI	8pt, Black
Data Labels	Segoe UI	9pt, Black, Bold
KPI Card Value	Segoe UI Semibold	24pt, Black

2. Data Sources

2.1 Loaded Tables

The following tables are loaded directly from the source system into the Power BI data model:

#	Table Name	Type
1	Patients	Fact Table (F)
2	Branches	Dimension Table (D)
3	Countries	Dimension Table (D)

2.2 Created / Reference Tables

The following dimension and reference tables were created within Power BI to enrich the data model:

2.2.1 Calendar Table (DimDate)

A date dimension table generated programmatically from the Patients table using the minimum and maximum values of the RegistrationDate column. It provides standard date hierarchies (Year, Quarter, Month Name, Day) to support time-intelligence analysis.

2.2.2 Blood Group Description Table

ID	Blood Group
0	O-
1	O+
2	A-
3	A+
4	B-
5	B+
6	AB-
7	AB+

2.2.3 Insurance Description Table

ID	Insurance Company
1	GULF UNION AL AHLIA COOPERATIVE INSURANCE
2	SAUDI ARABIAN COOPERATIVE INSURANCE CO. (SAICO)

3	BUPA Insurance Company
4	Medgulf Life
5	United Cooperative Assurance
6	Metlife AIG ANB Cooperative Insurance Company
7	Al-EtiHAD Co-operative Insurance Co - TCS
0	Cash

2.2.4 Source of Awareness Table

ID	Description
1	Doctor Referral
2	Medical Representative Referral
3	Police Referral
4	Ambulance Service
5	Hospital Website
6	Family / Friend Referral
7	Google Search
8	Walk-in
9	Other
10	Corporate Tie-up
11	Snapchat
12	LinkedIn
13	Instagram
14	Discord
15	Twitter

3. Data Transformations

All transformations are applied in Power Query (M language) before data is loaded into the model.

3.1 Branches Table

- Removed blank rows to ensure data integrity.
- Applied Title Case capitalization to the BranchNameEn column for display consistency.
- Removed unnecessary columns not relevant to the analytical model.

3.2 Countries Table

- Applied Title Case capitalization to the CountryNameEn column.
- Removed unnecessary columns not relevant to the analytical model.

3.3 Patients Table (Fact Table)

Multiple transformations were applied to improve readability and enable segmentation analysis:

3.3.1 Gender Column — Value Replacement

Original Value	Replaced With
1	Male
2	Female

3.3.2 Billing Type Column — Value Replacement

Original Value	Replaced With
0	Cash
1	Insurance
2	Charity

3.3.3 ResidencyType Column — Value Replacement

Original Value	Replaced With
1	Citizen
2	Resident

3	Visitor
---	---------

3.3.4 Calculated Column: Age

A calculated Age column is derived by subtracting the patient's date of birth (DOB column) from today's date, yielding the patient's current age in years.

3.3.5 Calculated Column: Age Group

An Age Group classification column is derived from the Age column using the following conditional logic:

Condition	Age Group Label
Age <= 2	Infants
Age <= 12	Children
Age <= 19	Teens
Age <= 35	Young Adults
Age <= 59	Adults
Age >= 60	Seniors

- Removed unnecessary columns not required by the analytical model.

4. Data Modelling

The data model follows a star schema design, with the Patients fact table at the centre connected to multiple dimension tables. All relationships use One-to-Many cardinality with Single cross-filter direction to optimise query performance.

From Table	From Column	To Table	To Column	Cardinality	Cross Filter
Branches	BranchId	Patients	BranchId	One-to-Many	Single
Countries	CountryId	Patients	Country	One-to-Many	Single
Blood Group Description	Id	Patients	Blood Group	One-to-Many	Single
Insurance Description	Id	Patients	InsuranceCompanyId	One-to-Many	Single
DimDate	Date	Patients	Registration Date	One-to-Many	Single
Source of Awareness	Id	Patients	Source_of_Awareness	One-to-Many	Single

5. Key Performance Indicators (Measures)

The following DAX measures are defined in the data model and used across all report pages:

Measure	DAX Expression	Description
Total Patients	<code>CALCULATE (COUNT (Patients [PatientID]))</code>	Count of all patients
Male Patients	<code>CALCULATE (COUNT (Patients [PatientID]), Patients [Gender] = "Male")</code>	Count of male patients
Female Patients	<code>CALCULATE (COUNT (Patients [PatientID]), Patients [Gender] = "Female")</code>	Count of female patients

6. Report Pages & Visualizations

6.1 Patient Demographic Insights Dashboard

6.1.1 Dashboard Header (Text Card)

Text Card Configuration

Property	Configuration
Title Text	Patient Demographic Insights
Title Font	Segoe UI, 18pt, Black, Bold
Subtitle Font	Segoe UI, 10pt, Black, Bold

6.1.2 Slicers

Branch Slicer

Property	Configuration
Size (H x W)	60 x 190 px
Position	1077 x 15 px
Field	Branches[BranchNameEn]
Header Font	DIN, 11pt, Black, Bold
Value Font	Segoe UI, 8pt, Black, Bold

Year Slicer

Property	Configuration
Size (H x W)	60 x 113 px
Position	1282 x 15 px
Field	DimDate[Year]
Header Font	DIN, 11pt, Black
Value Font	Segoe UI, 8pt, Black

6.1.3 KPI Card

Total Patients (Name Card)

Property	Configuration
Size (H x W)	75 x 134 px
Measure	Total Patients
Font	Segoe UI Semibold, 24pt, Black

6.1.4 Charts

Patients by Gender (Donut Chart)

Property	Configuration
Size (H x W)	350 x 450 px
Legend Field	Patients[Gender]
Values Field	Total Patients (measure)
Data Labels	On — Outside, All Detail Labels, Segoe UI 9pt Black
Title	DIN, 14pt, Black, Bold Italic
Subtitle	Segoe UI, 9pt, Black

Patients Trend (Line Chart)

Property	Configuration
Size (H x W)	350 x 915 px
X-Axis Fields	DimDate[Year], DimDate[Quarter], DimDate[Month Name]
Y-Axis Field	Total Patients (measure)
Axis Titles / Y Values	Off
Data Labels	On — Segoe UI 9pt Black Bold
Title	DIN, 14pt, Black, Bold Italic
Subtitle	Segoe UI, 9pt, Black

Patients Demographic / Residency (Donut Chart)

Property	Configuration
Size (H x W)	350 x 450 px
Legend Field	Patients[ResidenceTypeID]
Values Field	Total Patients (measure)
Data Labels	On — Outside, All Detail Labels, Segoe UI 9pt Black
Title	DIN, 14pt, Black, Bold Italic
Subtitle	Segoe UI, 9pt, Black

Patients by Age Group (Stacked Bar Chart)

Property	Configuration
Size (H x W)	350 x 450 px
X-Axis Field	Total Patients (measure)
Y-Axis Field	Patients[Age Group]
Tooltip	Male Patients, Female Patients
Y-Axis Values	On

Data Labels	On — Segoe UI 9pt Black Bold
Title	DIN, 14pt, Black, Bold Italic
Subtitle	Segoe UI, 9pt, Black
Header Icons / Tooltip	Enabled

Patients by Blood Group (Stacked Column Chart)

Property	Configuration
Size (H x W)	350 x 450 px
X-Axis Field	Blood Group Description[Description]
Y-Axis Field	Total Patients (measure)
Tooltip	Male Patients, Female Patients
X-Axis Values	On
Data Labels	On — Segoe UI 9pt Black Bold
Title	DIN, 14pt, Black, Bold Italic
Subtitle	Segoe UI, 9pt, Black
Header Icons / Tooltip	Enabled

How Do You Know About Us (Stacked Bar Chart)

Property	Configuration
Size (H x W)	350 x 450 px
X-Axis Field	Total Patients (measure)
Y-Axis Field	Source of Awareness[Description]
s	Male Patients, Female Patients
Y-Axis Values	On
Data Labels	On — Segoe UI 9pt Black Bold
Title	DIN, 14pt, Black, Bold Italic
Subtitle	Segoe UI, 9pt, Black
Header Icons / Tooltip	Enabled

Patients by Country (Stacked Bar Chart)

Property	Configuration
Size (H x W)	350 x 450 px
X-Axis Field	Total Patients (measure)
Y-Axis Field	Countries[CountryNameEN]
Tooltip	Male Patients, Female Patients
Y-Axis Values	On

Data Labels	On — Segoe UI 9pt Black Bold
Title	DIN, 14pt, Black, Bold Italic
Subtitle	Segoe UI, 9pt, Black
Header Icons / Tooltip	Enabled

Patients by Billing Segment (Donut Chart)

Property	Configuration
Size (H x W)	350 x 450 px
Legend Field	Patients[BillingType]
Values Field	Total Patients (measure)
Data Labels	On — Outside, All Detail Labels, Segoe UI 9pt Black
Title	DIN, 14pt, Black, Bold Italic
Subtitle	Segoe UI, 9pt, Black

6.2 Patients Registration Trend Dashboard

6.2.1 Dashboard Header (Text Card)

Text Card Configuration

Property	Configuration
Title Text	Patients Registration Trend
Title Font	Segoe UI, 18pt, Black, Bold
Subtitle Font	Segoe UI, 10pt, Black, Bold

6.2.2 Slicers

Year Slicer

Property	Configuration
Size (H x W)	60 x 113 px
Position	1077 x 15 px
Field	DimDate[Year]
Header Font	DIN, 11pt, Black, Bold
Value Font	Segoe UI, 8pt, Black, Bold

Branch Slicer

Property	Configuration
Size (H x W)	60 x 190 px
Position	1205 x 15 px
Field	Branches[BranchNameEn]
Header Font	DIN, 11pt, Black, Bold
Value Font	Segoe UI, 8pt, Black, Bold

Blood Group Slicer

Property	Configuration
Size (H x W)	60 x 113 px
Position	949 x 90 px
Field	Blood Group Description[Description]
Header Font	DIN, 11pt, Black, Bold
Value Font	Segoe UI, 8pt, Black, Bold

Gender Slicer

Property	Configuration
----------	---------------

Size (H x W)	60 x 113 px
Position	1077 x 90 px
Field	Patients[Gender]
Header Font	DIN, 11pt, Black, Bold
Value Font	Segoe UI, 8pt, Black, Bold

How Do You Know About Us Slicer

Property	Configuration
Size (H x W)	60 x 190 px
Position	1205 x 90 px
Field	Source of Awareness[Description]
Header Font	DIN, 11pt, Black, Bold
Value Font	Segoe UI, 8pt, Black, Bold

6.2.3 Charts

Annual Registration Trend (Stacked Column Chart)

Property	Configuration
Size (H x W)	350 x 682 px
X-Axis Field	DimDate[Year]
Y-Axis Field	Total Patients (measure)
X-Axis Values	On
Data Labels	On — Segoe UI 9pt Black Bold
Title	DIN, 14pt, Black, Bold Italic
Subtitle	Segoe UI, 9pt, Black
Header Icons / Tooltip	Enabled

Quarterly Registration Trend (Clustered Column Chart)

Property	Configuration
Size (H x W)	350 x 682 px
Legend Field	DimDate[Year]
X-Axis Field	DimDate[Quarter]
Y-Axis Field	Total Patients (measure)
X-Axis Values	On
Data Labels	On — Segoe UI 9pt Black Bold
Title	DIN, 14pt, Black, Bold Italic
Subtitle	Segoe UI, 9pt, Black

Header Icons / Tooltip	Enabled
------------------------	---------

Monthly Registration Trend (Clustered Column Chart)

Property	Configuration
Size (H x W)	350 x 1380 px
Legend Field	DimDate[Year]
X-Axis Field	DimDate[Month Name]
Y-Axis Field	Total Patients (measure)
X-Axis Values	On
Data Labels	On — Segoe UI 9pt Black Bold
Title	DIN, 14pt, Black, Bold Italic
Subtitle	Segoe UI, 9pt, Black
Header Icons / Tooltip	Enabled

Daily Registration Trend (Clustered Column Chart)

Property	Configuration
Size (H x W)	350 x 1380 px
X-Axis Fields	DimDate[Year], DimDate[Quarter], DimDate[Month Name], DimDate[Day]
Y-Axis Field	Total Patients (measure)
X-Axis Values	On
Data Labels	On — Segoe UI 9pt Black Bold
Title	DIN, 14pt, Black, Bold Italic
Subtitle	Segoe UI, 9pt, Black
Header Icons / Tooltip	Enabled